

Analysis of Trader Behavior and Market Sentiment Using Fear–Greed Index and Hyperliquid Trading Data

1. Introduction

Financial markets are strongly influenced by collective human emotions, especially during periods of uncertainty or hype. Traders often react not only to price movements but also to the psychological climate of the market. In cryptocurrency markets, this psychological state is frequently summarized by the *Fear and Greed Index*, which captures how pessimistic or optimistic the market is on a given day.

This project aims to understand how trader behavior—such as trade size, risk exposure, and profitability—varies during different sentiment conditions. By combining historical trading records with daily sentiment values, we can investigate whether traders behave more aggressively during greed phases, whether losses increase during high-risk periods, and how sentiment shapes trading outcomes. The results of this analysis provide meaningful insights for both human traders and algorithmic trading systems.

2. Dataset Description

The project is based on two separate datasets that were merged using their common date fields.

2.1 Sentiment Dataset

This dataset represents the Bitcoin market's daily emotional state and contains:

- **timestamp** – exact timestamp of market sentiment record
- **value** – numerical sentiment score between 0 and 100
- **classification** – categorical label (*Fear* or *Greed*)
- **date** – daily consolidated date value

The sentiment score provides a simple but powerful metric for understanding crowd psychology: Low values represent fear-driven markets, while high values indicate greed-driven optimism.

2.2 Hyperliquid Trader Dataset

This dataset contains real executed trades from various accounts. Key attributes include:

- Account
- Coin
- Execution Price
- Size Tokens and Size USD
- Side (Buy/Sell)
- Start Position and Direction
- Fee
- Closed PnL
- Timestamp IST and Timestamp

These fields allow us to analyze how much traders risk, how frequently they win or lose, and how behavior changes over time.

3. Data Preparation and Feature Engineering

The first step was to convert timestamps from both datasets into a standardized date format to enable accurate merging. Since the Hyperliquid dataset contained timestamps in IST, the date was extracted from these fields and aligned with the sentiment dataset.

Next, several useful numerical features were engineered:

3.1 Trade Volume

To capture the overall size of each trade:

$$\text{trade_volume} = \text{Size USD}$$

If Size USD was missing, it was computed as:

$$\text{trade_volume} = \text{Execution Price} \times \text{Size Tokens}$$

3.2 Risk Exposure

A simple measure of how much capital a trader puts at risk:

$$\text{risk_exposure} = \text{Execution Price} \times \text{Size Tokens}$$

3.3 Target Variables

Two types of targets were used:

- **Classification Target:** Whether a trade resulted in a loss (Closed PnL < 0)
- **Regression Target:** Predicting the actual Closed PnL of the trade

Once the data was cleaned and processed, it was saved as a unified dataset for modeling.

4. Methodology

To keep the analysis computationally efficient and easy to interpret, lightweight machine learning models were intentionally selected.

4.1 Classification Model

A **Logistic Regression** model was used to predict whether each trade would result in a loss. The following features were used:

- Trade Volume
- Risk Exposure
- Sentiment Score

Logistic Regression was chosen for its simplicity and interpretability. It allowed us to identify how each feature increased or decreased the probability of a loss.

4.2 Regression Model

A **Random Forest Regressor** with only 20 trees was used to estimate the Closed PnL value. This is a low-complexity model but still strong enough to capture non-linear relationships.

4.3 Evaluation

Model performance was assessed using:

- **Classification:** Precision, Recall, F1-score, Confusion Matrix
- **Regression:** RMSE (Root Mean Squared Error), Actual vs Predicted Plot
- **Feature Importance:** Identifying which features contributed most to predictions

The focus of the modeling was insight generation, not perfect prediction.

5. Results and Findings

5.1 Behavior Patterns During Fear vs. Greed

The merged dataset revealed clear behavioral shifts:

- During **Greed**, traders tend to increase their trade sizes and risk exposure.
- During **Fear**, trades are smaller, more cautious, and result in more stable outcomes.

This confirms that market psychology influences trading decisions significantly.

5.2 Prediction of Losses

The Logistic Regression model showed:

- Higher risk exposure strongly increases the probability of a loss.
- Large trades are more likely to result in negative outcomes.
- Sentiment score slightly moderates this behavior, with greed conditions amplifying risk-taking.

Even a simple model demonstrated that loss events are **not random** but consistently linked with aggressive trading patterns.

5.3 PnL Regression Insights

While predicting exact PnL is difficult due to market volatility, the regression model revealed:

- **Risk exposure** is the most influential factor in determining PnL.
- **Trade volume** also plays a significant role.
- **Sentiment value** affects PnL variability — greed periods show wider profit/loss swings.

This suggests that traders who adjust their risk levels based on sentiment could reduce unnecessary losses.

6. Discussion

The analysis highlights important relationships between sentiment and trading behavior:

- Traders tend to follow market emotion rather than objective risk-management rules.
- Many losses stem from periods of overconfidence, especially when sentiment is high.
- Even simple quantitative models can capture meaningful behavioral trends.
- Market sentiment should be considered a **risk factor**, not just an indicator of opportunity.

These insights can help trading firms enhance risk models, guide traders to avoid emotional decisions, and support algorithmic strategies that adapt to sentiment shifts.

7. Conclusion

This project successfully demonstrates how combining trader activity with market sentiment provides a deeper understanding of trading behavior. By enriching trade-level data with sentiment context, we uncover patterns that highlight the psychological side of trading.

Although the machine learning models were intentionally simple, they revealed which factors drive losses and how risk-taking changes between fear and greed states. The results emphasize the importance of sentiment-aware trading strategies and reinforce the idea that disciplined risk management is crucial, especially in highly emotional markets.

Future work could include:

- Adding leverage and margin data
- Using time-series or sequential deep learning models
- Incorporating technical indicators or volatility measures
- Building a sentiment-adaptive trading policy for backtesting

This analysis forms a strong foundation for sentiment-driven trading insights in Web3 markets.